



Multiple Choice:

1. If a resistance to which a constant voltage is applied is halved, what power dissipation will result?

A. Doubled
B. Halved
C. Stays the same
D. Quadrupled

2. If the reference point is changed to the earth point then any voltage taken w.r.t. the earth is known as

A. Absolute potential
B. Earth ground
C. Relative voltage
D. Absolute point

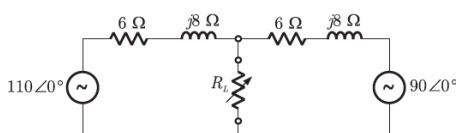
3. What is the significance of connecting loads in parallel?

A. It makes power consumption less
B. It provides greater efficiency
C. It increases the safety factor
D. It allows independent operations of loads

4. Which of the following safety rules should be observed while working on live electric circuit?

A. Keep yourself well insulated from earth ground.
B. When making measurements in a live circuit place one hand behind your back or in your pocket.
C. Make resistance measurements only in a live circuit.
D. Both A and B.

5. Two ac sources feed a common variable resistive load as shown in figure. Under the maximum power transfer condition, what is the power absorbed by the load resistance R_L ?



A. 526 W
B. 562 W
C. 625 W
D. 652 W

6. The hot resistance of an incandescent lamp is about _____ times its cold resistance.

A. 10
B. 50
C. 5
D. 100

7. If a resistor in a series circuit is shorted, the series current, I ,

A. decreases
B. stays the same.
C. increases.
D. drops to zero.

8. The equivalent wye element of 3 equal resistors each equal to R and connected in delta is

A. R
B. $R/3$
C. $3/2 R$
D. $3R$

9. A coil of aluminum wire has a resistance of 50Ω when its temperature is 0°C . Determine its resistance at 100°C if the temperature coefficient of resistance of aluminum at 0°C is $0.0038/^\circ\text{C}$.

A. 69Ω
B. 65Ω
C. 96Ω
D. 90Ω

10. A certain factory takes 250 A at 220 V. It is supplied through a pair of feeders 73.2 meters long and 1.29 cm^2 in cross section. What voltage must be maintained at the supply end of the feeders? (Use $\rho = 1.78 \mu\Omega\text{-cm}$)

A. 111.5 V
B. 222.5 V
C. 333.5 V
D. 444.5 V

11. A wire with a smaller cross-sectional area will produce:

A. less heat
B. more conductance
C. less resistance
D. more heat

12. Halving the cross-sectional area of a conductor will:

A. Not affect the resistance.
B. Quarter the resistance.
C. Half the resistance.
D. Double the resistance.

13. An aluminum wire has a resistance of 50Ω at 20°C . The wire is heated to a temperature of 100°C . Determine the resistance of the wire at 100°C , assuming that the temperature coefficient of resistance at 0°C is $0.004/^\circ\text{C}$

A. 64.81Ω
B. 61.84Ω
C. 68.14Ω
D. 63.16Ω

14. In Norton's Theorem, the short circuit current is obtained by

A. opening the load terminals
B. shorting the load terminals
C. opening the voltage source
D. shorting the voltage source

15. Delta to wye or wye to delta transformation technique is applied to a _____ network.

A. one-terminal
B. two-terminal
C. three-terminal
D. complex

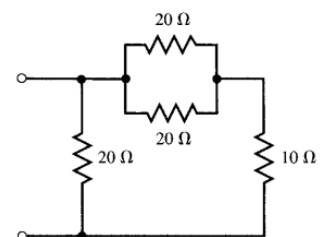
16. If the diameter of a wire is doubled, its current carrying capacity becomes

A. one-fourth
B. half
C. twice
D. four times

17. If a resistance element is part of two loops, how many voltage drops must be calculated for that component?

A. Two
B. None
C. One
D. Three

18. Find the equivalent resistance of the figure shown below:



A. 10 ohms
B. 20 ohms
C. 15 ohms
D. 5 ohms



19. Suppose you double the voltage in a simple dc circuit and cut the resistance in half. The current will become:
- A. Four times as great
 - B. Twice as great
 - C. The same as it was before
 - D. Half as great
20. In a series circuit, the current, I , is
- A. different in each resistor.
 - B. the same everywhere.
 - C. the highest near the positive and negative terminals of the voltage source.
 - D. different at all, the largest resistance has
21. When a material becomes a superconductor, its resistivity becomes
- A. very small
 - B. zero
 - C. about 10% of normal value
 - D. about 20% of normal value
22. "The sum of the emf's and potential differences around a closed loop equals zero" is a consequence of:
- A. Newton's third law
 - B. Newton's second law
 - C. Conservation of energy
 - D. Conservation of charge
23. Two (2) 115-V incandescent lamps A and B are connected in series across a 230-V source. If lamp A is rated 75 watts and lamp B is rated 50 watts, determine the current drawn by the series connection.
- A. 0.52 A
 - B. 0.48 A
 - C. 0.64 A
 - D. 0.57 A
24. A network contains linear resistors and ideal voltage sources. If values of all the resistors are doubled, then the voltage across each resistor is
- A. increase by 4 times
 - B. half
 - C. doubled
 - D. no change
25. Which of the following is neither a basic physical law nor derivable from one?
- A. Ohm's law
 - B. Coulomb's law
 - C. Kirchhoff's first law
 - D. Kirchhoff's second law
26. Which of the following is the poorest conductor of electricity?
- A. Aluminum
 - B. Nichrome
 - C. Copper
 - D. All of the above
27. If a resistor in a series circuit becomes open, how much is the voltage across each of the remaining resistors that are still good?
- A. Each good resistor has the full value of applied voltage.
 - B. The applied voltage is split evenly among the good resistors.
 - C. 0 V.
 - D. It cannot be determined.
28. A series circuit in which desired portions of the source voltage may be tapped off for use in equipment.
- A. Voltage selector
 - B. Dividing network
 - C. Voltage trap
 - D. Voltage divider
29. If a complex circuit is reduced to an equivalent circuit consisting of a single voltage source in series with a single resistor, this is an example of:
- A. Norton's theorem
 - B. Thevenin's theorem
 - C. Ohm's law
 - D. Kirchhoff's law
30. If one branch of a parallel circuit is short-circuited.
- A. the fuse in the main line will blow.
 - B. the voltage across the short-circuited branch will measure the full value of applied voltage.
 - C. all the remaining branches are effectively short-circuited as well.
 - D. Both A and C.
31. Twelve $1\ \Omega$ resistance are used as edges to form a cube. What is the resistance between two diagonally opposite corners of the cube?
- A. $5/6\ \Omega$
 - B. $6/5\ \Omega$
 - C. $1\ \Omega$
 - D. $3/2\ \Omega$
32. The polarity of a resistor's voltage drop is determined by
- A. the direction of current through the resistor.
 - B. how large the resistance is.
 - C. how close the resistor is to the voltage source.
 - D. how far away the resistor is from the voltage source
33. The electric energy required to raise the temperature of a given amount of water is 1000 kWh. If heat losses are 25%, the total heating energy required is ____ kWh.
- A. 1500
 - B. 1333
 - C. 1250
 - D. 1000
34. The resistance of a 5 m length of wire is $600\ \Omega$. Determine the resistance of an 8 m length of the same wire.
- A. $960\ \Omega$
 - B. $906\ \Omega$
 - C. $690\ \Omega$
 - D. $609\ \Omega$
35. What is expected when two $20\ \text{k}\Omega$, 1 watt resistor in parallel are used instead of one $10\ \text{k}\Omega$, 1 watt?
- A. Provides higher current
 - B. Provides less power
 - C. Provides more power
 - D. Provides wider tolerance
36. Twelve identical wires of resistance 6 ohms each are arranged to form the edge of a cube. A current of 40 mA is led into the cube at one corner and out at the other diagonally opposite corner. Calculate the potential difference developed between these corners.



- A. 0.20 V C. 0.28 V
B. 0.22 V D. 0.24 V
37. The superposition theorem is used when the circuit contains
- A. reactive elements
B. active elements
C. number of voltage sources
D. single voltage source
38. In a normally operating parallel circuit, the individual branch currents are
- A. independent of each other.
B. not affected by the value of the applied voltage.
C. larger than the total current, I_T .
D. None of the above.
39. What is a closed path made of several branches of the network called?
- A. Junction C. Node
B. Terminal D. Loop
40. What is expected when two 20 k Ω , 1 watt resistor in parallel are used instead of one 10k Ω , 1 watt?
- A. Provides higher current
B. Provides less power
C. Provides more power
D. Provides wider tolerance
41. If one branch of a parallel circuit becomes open,
- A. all remaining branch currents increase.
B. the voltage across the open branch will be 0 V.
C. the remaining branch currents do not change in value.
D. the equivalent resistance of the circuit decreases.
42. In a parallel circuit, the individual branch currents are
- A. not related to the branch resistance values
B. directly proportional to the branch resistance values
C. inversely proportional to the branch resistance values.
- D. None of the above
43. In any network made up of linear resistances and containing more than one source of e.m.f., the resultant current flowing in any branch is the algebraic sum of the currents that would flow in that branch if each source was considered separately. What is this theorem?
- A. Thevenin's theorem
B. Norton's theorem
C. Superposition theorem
D. Millman's theorem
44. Two resistors of 4 ohms and 6 ohms are connected in parallel. If the total current is 30 amperes, find the current through each.
- A. 17 A and 13 A
B. 18 A and 12 A
C. 20 A and 30 A
D. 16 A and 14 A
45. If an emf in circuit A produces a current in circuit B, then the same emf in circuit B produces the same current in circuit A. This theorem is known as
- A. Maximum power transfer theorem
B. Millman's theorem
C. Reciprocity theorem
D. Norton's theorem
46. A 100- Ω resistor, R1, and a 300- Ω resistor, R2, are in parallel across a dc voltage source. Which resistor dissipates more power?
- A. The 300- Ω resistor.
B. Both resistors dissipate the same amount of power.
C. The 100- Ω resistor.
D. It cannot be determined.
47. The power transferred from a supply source to a load is at its maximum when the resistance of the load is _____ the internal resistance of the source.
- A. lower than
B. greater than
- C. equal to
D. one-half of the
48. What If you have a 12-strand conductor and each strand has a radius (one-half the diameter) of 2/10 inch, what is the square mil area of the conductor?
- A. 1507965 sq. mil
B. 1705965 sq. mil
C. 1607955 sq. mil
D. 1907565 sq. mil
49. A 5- Ω and 10- Ω resistor are connected in series across a dc voltage source. Which resistor will dissipate more power?
- A. the 5- Ω resistor.
B. the 10- Ω resistor.
C. It depends on how much the current is.
D. They will both dissipate the same amount or power.
50. The first step in Thevenin's theorem is to:
- A. determine the short-circuit current
B. open the load under test
C. short the load under test
D. determine the open-circuit current
- END